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Surgical placement of the vagus nerve stimulator

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KEYWORDS:
Seizures;
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Biomedical device

The vagus nerve stimulator has become increasingly more common as an adjunctive therapy for patients with antiepileptic medication resistance. The surgical procedure is performed within the left cervical portion of the vagus nerve with the combined pulse generator/battery placed in a preaxillary position. A growing number of otolaryngologists are performing this procedure. This article is presented to assist in the surgeon's understanding of the device parts, patient/guardian consent requirements, preoperative planning on the part of the surgeon as well as the operating room team, intraoperative surgical techniques to maximize the success of the surgery, and methods for decreasing the risk of complications.

The vagus nerve stimulator (VNS) therapy system, produced by Cyberonics, Inc (Houston, TX), includes an FDA-approved medical device that is placed on the cervical vagus nerve to treat patients with seizures recalcitrant to antiepileptic medications (Figure 1). The number of persons benefiting from this procedure continues to increase. As of January 2004, more than 25,000 patients have been implanted with the VNS therapy device.[1] If the U.S. FDA approves VNS therapy for treatment-resistant depression, the need for trained surgeons to perform the procedure could increase exponentially. Currently, the majority of VNS therapy devices are placed by neurosurgeons. It is the contention of this author/otolaryngologist that this procedure should be performed by surgeons who are trained in neck and cranial nerve procedures as well as microscopic and cosmetic techniques. In addition, a surgeon who is trained in laryngeal examination can evaluate any potential true vocal cord immobility preoperatively or postoperatively, because left true vocal cord immobility is a potential complication of this procedure.[2] This article will acquaint the otolaryngologist with surgical implantation of the VNS therapy device and assist in the peri-, intra-, and postoperative evaluation and management of the patient.

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History of vagus nerve stimulation

The theoretical rationale of afferent stimulation was considered 2,000 years ago by Pelops, the master of Galen. The physician would apply a painful stimuli to a limb with ligatures to abort a focal seizure.[3] In 1938, a cat model was developed whereby afferent vagal stimulation was noted to affect electroencephalogram activity.[4] Similar contemporary studies have shown that timed stimulation of the vagus nerve can abort seizures in multiple animal models.[5] The actual mechanism of how vagus nerve stimulation suppresses seizures is unknown. However, the antiseizure effects of vagus nerve stimulation are likely the result of altered activity among the systems associated with the diffuse anatomic distribution of vagal projections, including the reticular activating system, central autonomic network, limbic system, and noradrenergic projection system.[6]

Indications

The FDA approved the use of the VNS therapy system "as an adjunctive therapy in reducing the frequency of seizures in adults and adolescents over 12 years of age with partial onset seizures that are refractory to antiepileptic medications."[7] Neurologists have extended the use of the device for seizures other than partial onset[8,9]and have prescribed it for infants and children.[10,11] In addition, the manufacturer has submitted an application to the U.S. FDA to obtain approval for use of the device in the treatment of depression.[12] The procedure is potentially reversible for the minority of patients who do not respond or who require explantation because of infection.[13]

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Figure 1 Photograph of the VNS pulse generator and lead.

Contraindications

The VNS therapy device cannot be used in patients who have previously undergone left cervical vagotomy. Shortwave diathermy, microwave diathermy, or therapeutic ultrasound diathermy are contraindicated for patients implanted with the VNS therapy device. Diagnostic ultrasound is not considered harmful.[7] Among the warnings listed in the VNS therapy system physician's manual are dysphagia with aspiration risk[14]and obstructive sleep apnea[15] (the number of apneic episodes may increase with stimulation). A precaution listed in the physician's manual focuses on magnetic resonance imaging. Patients should not have magnetic resonance imaging with a body coil in the transmit mode. However, magnetic resonance imaging can be performed safely with a transmit-and-receive type of head coil with static magnetic field strength <2.0 Tesla and with the VNS therapy pulse generator turned off.[7] A relative contraindication is the existence of right vocal cord paralysis with a midline resting position, because the stimulation of the left side could cause transient stridor during the periods of stimulation, which usually last for 30 seconds.

Vagus nerve anatomy

Approximately 80% of the vagus nerve is composed of afferent fibers arising from the viscera, and less than 20% of the vagus nerve is composed of efferent fibers that innervate the larynx and maintain parasympathetic control of the heart, lungs, and gastrointestinal tract.[16] The segment of the cervical vagus nerve that receives stimulation from the VNS device is lateral and level to the mid laryngeal skeleton. This exposure provides for anatomical consistency of the vagus nerve within the carotid sheath between and posterior to the common carotid artery and the internal jugular vein. The manufacturer emphasizes the need to avoid the superior and inferior cervical cardiac branches of the vagus nerve.[7] These branches join fibers with the sympathetic trunk cardiac branches high within the neck, branch off from the vagus high in the neck (although the communications of the branching patterns are quite variable[17,18]), and innervate the cardiac plexus. The potential concern from the surgeon's point of view is the possibility of inadvertently placing the electrodes on the superior or inferior cardiac branches of the vagus nerve. The relatively small size and location of the cardiac branches outside the cardiac sheath reduce this possibility. The author has not experienced any difficulty with these branches.

Surgical technique

VNS surgical procedure for the left vagus nerve

Consent

Preoperative counseling includes obtaining consent for the procedure and explaining the attendant risks, benefits, and alternatives to the patient or guardian. The patient education videotape, available on request from the manufacturer, can be provided to the patient or the guardian. The surgeon should discuss with the patient or guardian the aspects of the procedure that create exposure for placement of the electrode and pulse generator, the local temporary discomfort anticipated, and the long-term loss of sensation in the distribution of the skin incisions. The patient should be told of the scar on the neck (the incision is placed within the Langer's lines and generally results in minimal scarring) and preaxillary region (a more obvious scar that will typically widen over time), as well as the palpable subcutaneous VNS therapy pulse generator and the occasionally palpable subcutaneous lead extending from the pulse generator to the electrodes. The patient should also be told that the voice might become hoarse while the device is in the stimulation cycle, usually programmed for 30 seconds. If the timing of stimulation and the accompanying induced hoarseness are inconvenient, the patient can temporarily abort the cycle by placing the VNS therapy magnet over the implanted pulse generator. The patient or guardian should be informed that the pulse generator has a limited battery life and will require replacement in 6 to11 years, depending on stimulation parameters. Other considerations include the possibilities that the patient will not respond to VNS therapy and that the lead could fail and require replacement with the attendant risk of vagus nerve injury.

Potential complications

Potential complications include infection, left true vocal cord paralysis or transient or permanent paresis, bleeding, seroma/hematoma, nausea, paresthesia, and bradycardia and ventricular asystole;[19] the estimated rate of asystole is 1 in 400 to 1 in 800 on the basis of open-label reporting.[1] No intraoperative asystole was reported during the clinical trials of VNS therapy.[19]

Side effects related to stimulation cycle

Side effects include coughing, laryngismus, voice alteration, local strap or deep cervical muscle movements or twitching, dyspnea, and local pain in the vicinity of the device.

Avoidance of complications

Infection

The risk of infection after the procedure is very low. During the clinical trials (N = 454), 1.1% of the patients developed infections that required explantation of the device, and 1.8% developed infections that did not require explantation.[1] Smaller studies, however, report infection rates from 3% to 8%.[20-22] When infection does occur, it is potentially catastrophic because it frequently requires explantation of the device. The risk of infection can be reduced by confirming that no other ongoing chronic infections such as occult sinusitis, tonsillitis, urinary tract infections, etc, are present. Such infections should be managed medically or surgically before exposing a foreign body (the VNS therapy device) to potential contamination from circulating bacteria.

Perioperative precautions to decrease the likelihood of infection include

1. Obtain a white blood cell count to evaluate for the presence of an occult infection and order an albumin serum level to rule out a catabolic state in patients with risk factors for infection or poor nutrition, respectively.
2. Have the patient bathe or shower, including the axilla and scalp, the night before the procedure.
3. Administer perioperative prophylactic antibiotics that will cover *Staphylococcus aureus*, intravenously before the procedure and orally for 3 days afterward.
4. Have the entire operating room staff give careful attention to adhering to sterile technique with regard to the patient as well as to the implantable devices.
5. Irrigate copiously the two surgical sites during the procedure and just before closure.
6. Carefully handle tissues, ie, avoid excessive cautery, tissue injury.
7. Minimize potential dead space of surgical access to the vagus nerve and the preaxillary space through conservative exposure and layered wound closure (drains not used).
8. Apply antibiotic ointment to the incisions for 2 to 3 days.
9. Avoid lifting the patient under the left axilla to prevent pulling of the incision site.

Left true vocal cord paralysis or paresis, transient or permanent

During the clinical trials of VNS therapy, vocal cord paralysis was reported in 0.7% of patients (N = 454). The affected patients improved overtime and regained full function.[23] Because most articles describing the clinical trials of VNS therapy do not document true vocal cord mobility, the incidence of paralysis and paresis is likely underreported.[2] The risk can be minimized by carefully handling the vagus nerve. Specifically, the vagus nerve should not be stretched, retracted with vascular loops, or devascularized with aggressive skeletonization of the nerve (ie, removal of the epineurium). This precaution includes identifying the proper size of electrode to place on the nerve. Two electrode sizes are available. Too small an electrode will cause a compression injury with Wallerian degeneration and paresis/paralysis.[24] Measuring the diameter of the vagus nerve before selecting the size of the electrode can help avoid vocal cord paralysis and paresis.[2]

Nursing considerations

Equipment

Room set up:

- Operating room table with left arm board
- Back table
- Mayo stand
- Unipolar cautery unit
- Bipolar cautery unit
- Small back table, nonsterile for laptop computer and programming wand
- Laser arm drape bag, for sterility of programming wand on sterile field
- Split sheet, arm sheet, head sheet
- Asepto syringe (C.R. Bard, Inc, Covington, GA) for irrigation

VNS therapy system components

Implanted components

- Model 102 Pulse Generator (Figure 1); we recommend having a sterile packaged backup for the pulse generator in case of loss of sterility
- Model 302 Bipolar Leads: 302-20 (2-mm diameter) and 302-30 (3-mm diameter) (Figure 2); we recommend having a sterile packaged backup for each lead in case of loss of sterility

Programming system components

- Computer or handheld computer (furnished by manufacturer)
- Model 201 Programming Wand
- Model 250 Programming Software

Component to be given to the patient

- Model 220 Magnet

Disposable components

- Model 402 Tunneler
- Model 502 Accessory Pack (ie, test resistor, extra lead tie-downs, and a hex screwdriver); the accessory pack provides extra items in the event that sterility is compromised

Special surgical instrumentation

- Standard otolaryngology head and neck soft tissue instrument tray

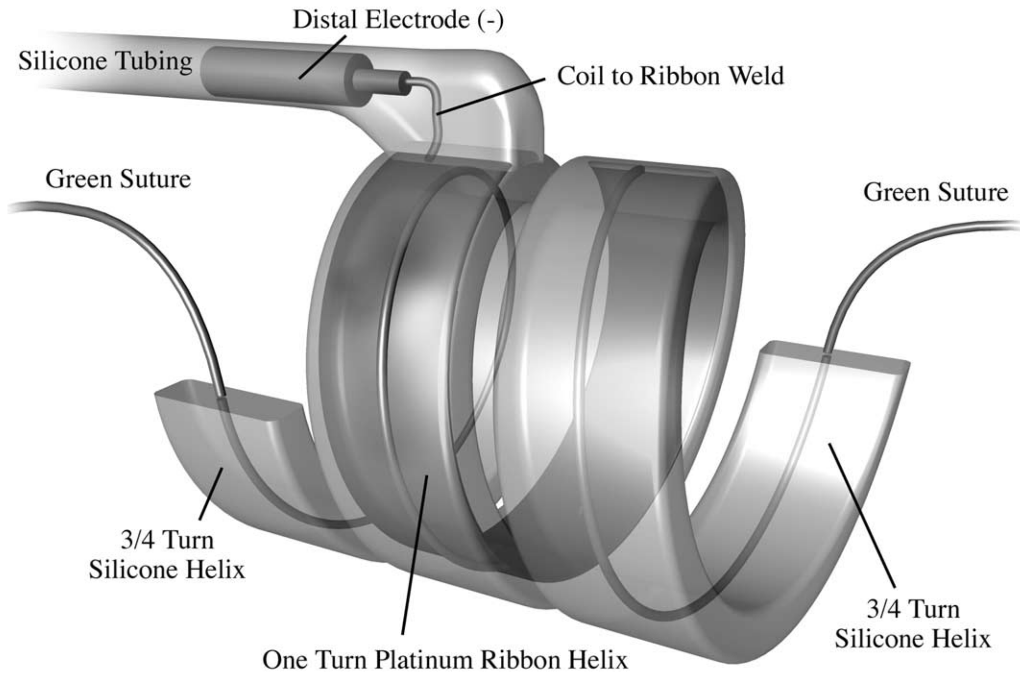

Figure 2 Figure of distal helical electrode (negative polarity, −) with embedded platinum iridium electrode ribbon and green grasping suture.

- Adjustable Weitlander Retractors (2) (Jarit Instruments Corp, Hawthorne, NY, Catalog Number 205-181) (Figure 3)
- Alm Retractor (Jarit Instruments Corp, Catalog Number 205-100) (Figure 3)
- Microvascular forceps, #8 (45° angled tip, 5-1/2) and #9 (straight tip) (Accurate Surgical & Scientific Instruments Corp, Westbury, NY, Catalog Numbers JFAL-3, JF-3-15)
- Calipers (20 mm; see note #5) (Codman Instruments Corp, Raynham, MA, Catalog Number 39-6500) (Figure 3)
- Surgical loupes or microscope may be helpful

Medications

- Antibiotic prophylaxis preoperatively (for *S. aureus* coverage)

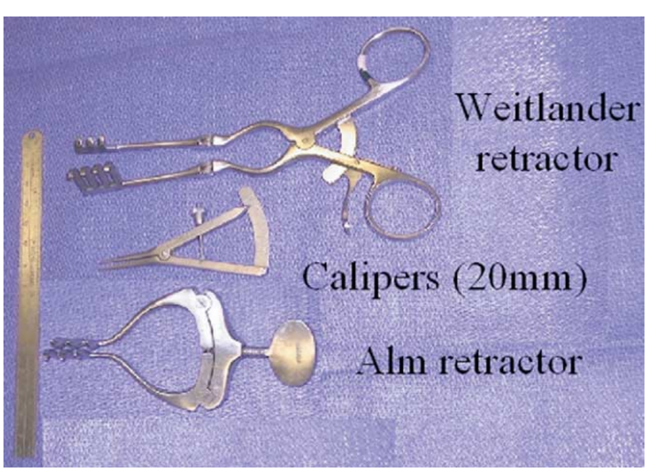

Figure 3 Photograph of instruments helpful for the dissection, exposure, and measurement of the vagus nerve. Top, Weitlander retractor; center, calipers (20 mm); bottom, Alm retractor.

- 1:200,000 epinephrine with or without lidocaine
- Antibiotic ointment

Patient positioning, prep, and drape

The patient should be positioned with head supported and rotated to the right with mild extension; the left arm should be secured to the arm board in a comfortable fashion abducted to 70° to 90° (Figure 4).

- Dermal infiltration of the planned surgical sites with epinephrine solution
- Standard prep, 10% providone iodine from lips to left ear to left elbow to posterior axilla to nipple to right neck to lips
- Drape and square off the prepped borders with towels and secure with staples; place drapes for wide exposure to maintain surgical landmarks

Special considerations

The programming wand will require a sterile plastic sleeve (eg, laser arm drape bag) for lead testing in the surgical field. The VNS therapy electrode packaging should not be opened until the diameter of vagus nerve has been measured and the appropriately sized electrode has been selected (2- or 3-mm inner diameter electrode). The accessory pack is available in the event that the sterility of one of the components, such as a lead tie-down or hex screwdriver, is compromised. The implant and warranty registration card should be completed at the time of surgery and returned to the manufacturer. The magnet is sent home with the patient. The referring neurologist should wait at least 2 weeks after the procedure to begin stimulation

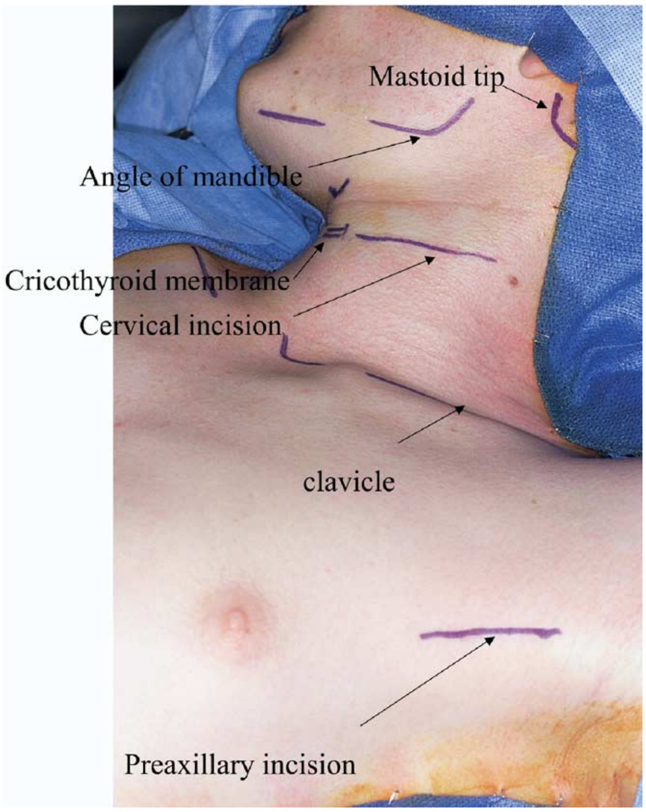

Figure 4 Photograph of patient positioning, draping, surgical landmarks, and planned cervical and preaxillary incisions.

Operative procedure: Part 1. Placement of the VNS therapy lead

The patient is placed under general anesthesia and may be paralyzed because the procedure does not require stimulation of any nerves. The planned incision sites are infiltrated with epinephrine solution after wiping the sites with alcohol. Sterile prepping and dressing technique and a prophylactic antibiotic are used, and the planned skin incisions are inked in the left anterior neck and preaxillary space (Figure 4). The planned neck incision is a curvilinear 5-cm-long incision made in a preexisting neck crease at the level of the mid to lower larynx between the lateral laryngeal skeleton and the sternocleidomastoid muscle. The planned preaxillary incision is a 5-cm-long incision placed along the lateral edge of the pectoralis major muscle in the preaxillary region. Within the pediatric population, the preaxillary incision can be made posterior to the lateral border of the pectoralis major so as to avoid the pulse generator from placing pressure on the incision line.[21] Other major anatomical landmarks are inked for orientation.

Vagus nerve exposure

Subplatysmal flaps are elevated 2 cm superiorly and inferiorly and retracted with the Weitlander retractor. The sternocleidomastoid muscle is dissected along its medial aspect and retracted posteriorly, and the strap muscles are retracted anteriorly with the laryngeal skeleton with a second Weitlander retractor. The anterior belly of the omohyoid muscle should be anterior and inferior to the route of dissection. The common carotid artery is palpated. Further sharp and blunt dissection along the medial aspect of the sternocleidomastoid muscle reveals the neurovascular sheath. The jugular vein is exposed and the branches are double-clip ligated or cauterized with the bipolar cautery. The common carotid adventitia is retracted anteriorly and the jugular vein adventitia is retracted posteriorly with the Alm retractor (Figure 5). The cervical vagus nerve is identified between the two vessels and freed from surrounding tissues for 3 cm with tenotomy scissors. The vagus nerve adventitia and vasa nervorum (blood supply surrounding the nerve) as well as all feeding blood vessels to the vagus nerve are preserved to the greatest extent possible. Nerve

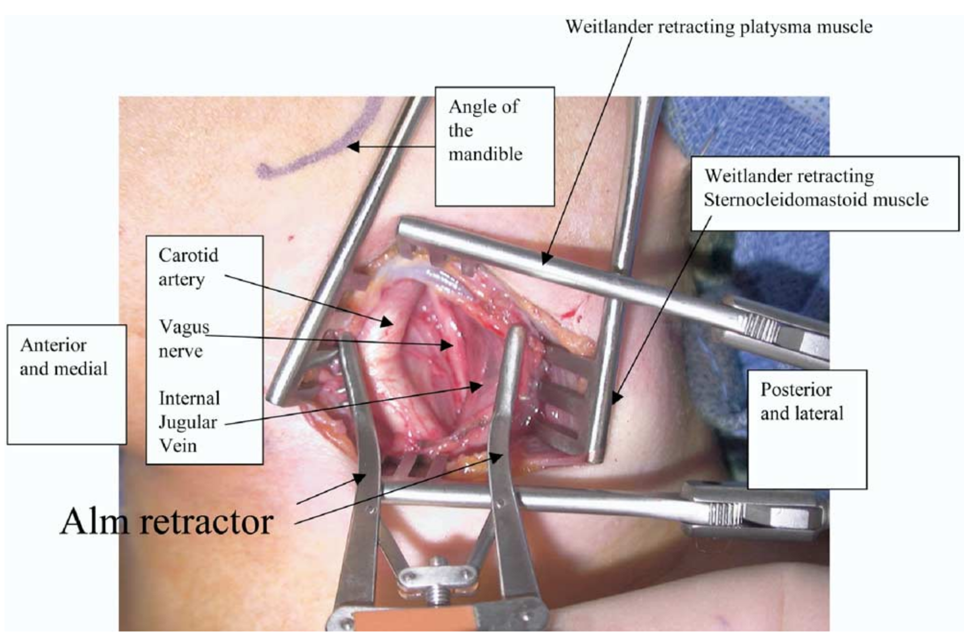

Figure 5 Photograph of surgical exposure of the vagus nerve with self-retaining retractors: (1) vertical set of Weitlander retractors separating the superficial tissues superiorly and inferiorly; (2) horizontal set of Weitlander retractors separating the sternocleidomastoid muscle and strap muscles posteriorly and anteriorly; and (3) Alm retractor separating the major vessels with exposure of the vagus nerve

crushing, retraction, and dehydration should be carefully avoided. The nerve epineurium is handled with microvascular forceps #8 (45° angled tip) and #9 (straight tip). Infrequently, depending on the relative position of the vagus nerve to the great vessels, it may be easier to retract both vessels anteriorly or posteriorly. The goal of retracting the vessels is to maximize exposure of the vagus nerve without placing traction on the nerve or occluding the jugular vein or the carotid artery.

Electrode placement

The vagus nerve diameter is measured with 20-mm calipers to allow for proper electrode size selection (Figure 6). A vagus nerve less than 2 mm in diameter receives a VNS therapy lead Model 302-20 (2-mm inner diameter helical electrode bipolar array). A vagus nerve greater than 2 mm in diameter receives a VNS therapy lead Model 302-30 (3-mm inner diameter helical electrode bipolar array). Once the properly sized electrode has been selected and the sterile inside packaging has been placed on the surgical field, the VNS therapy lead should be left in the sterile tray until needed. Do not immerse the electrode lead in saline or water before placement. Exposing the lead to fluid before placement can result in swelling of the insulated portions of the connector pin.

It is recommended that a surgeon unfamiliar with placing the electrode practice with dummy electrodes and a plastic "vagus nerve" before the first surgical procedure. A practice VNS therapy pulse generator and plastic "vagus nerve" is provided to the surgeon by the manufacturer. This practice will allow the surgeon to develop a sense of how to efficiently and atraumatically place the electrode around the vagus nerve. It will also provide the surgeon with the opportunity to feel how much retraction is required to unwind and recoil the electrode around the vagus nerve. The platinum iridium ribbon within the helical silicone electrode array is very delicate. Overstretching or crushing the platinum iridium ribbon is possible with the microvascular forceps. The electrodes are most safely handled by using the

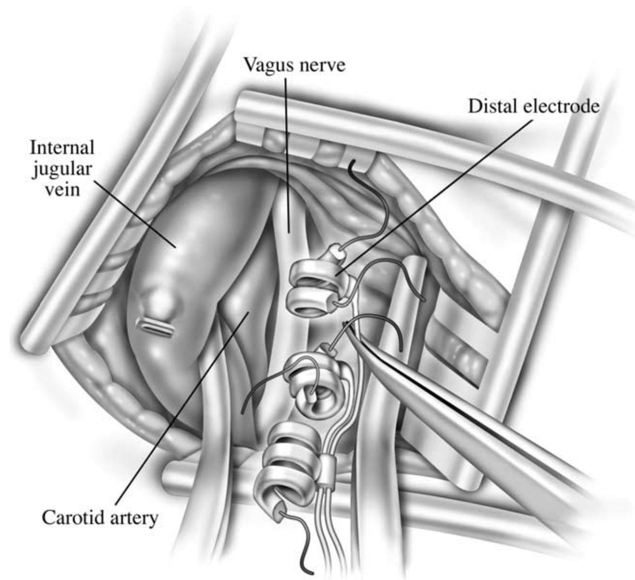

Figure 7 Helical electrodes positioned parallel to the vagus nerve.

embedded suture on either end of the electrode arrays and not handling the actual electrode array.

Although the technique for placing the electrode array may vary among surgeons, the author prefers the following technique. The distal electrode (ie, most cephalad) (negative polarity, −) is placed first and within the midportion of the vagus nerve. After the electrode is placed, it is slid in a cranial fashion to allow adequate exposure of the vagus nerve for the remaining middle electrode (positive polarity, +), which has white embedded sutures. The last helix to be placed resembles an electrode, but serves only as an "anchor tether." The electrodes on the VNS therapy lead are initially rested lateral and parallel to the vagus nerve (Figure 7). Next, the distal electrode (−) is wrapped around the vagus nerve by passing the microvascular forceps under the vagus nerve and grasping the most cephalad green embedded

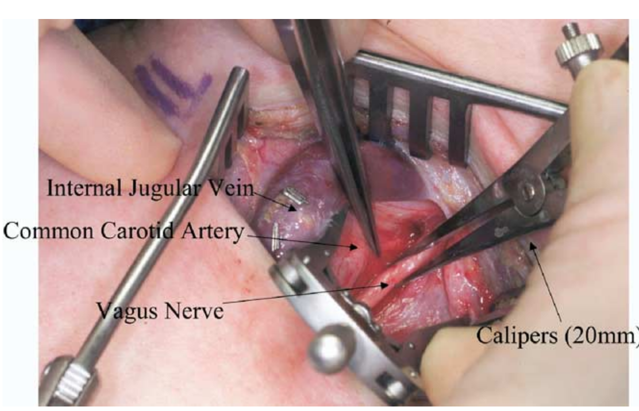

Figure 6 Photograph of the measurement of the diameter of the vagus nerve with calipers. This example shows both the internal jugular vein and the carotid artery retracted medially. (Reprinted from Santos PM: Evaluation of laryngeal function after implantation of the vagus nerve stimulation device. Otolaryngol Head Neck Surg 129:269-273, 2003. © 2003 with permission from the American Academy of Otolaryngology—Head and Neck Surgery Foundation, Inc.)

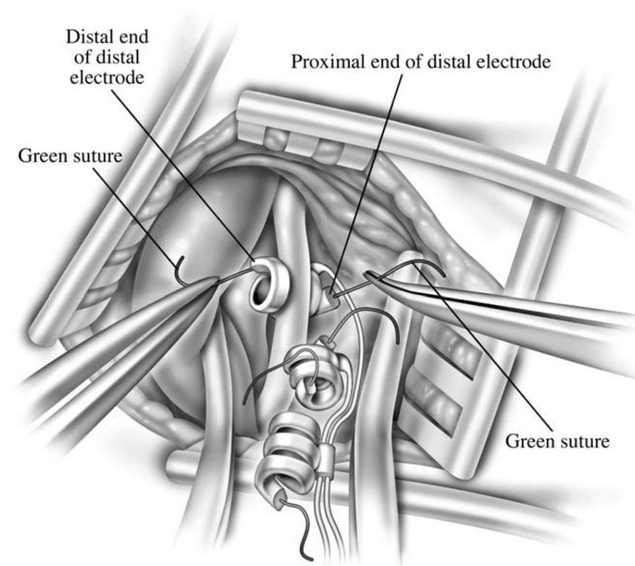

Figure 8 Initial placement of the distal spiral electrode (negative polarity, −) under the vagus nerve within the midportion of the helix (see text).

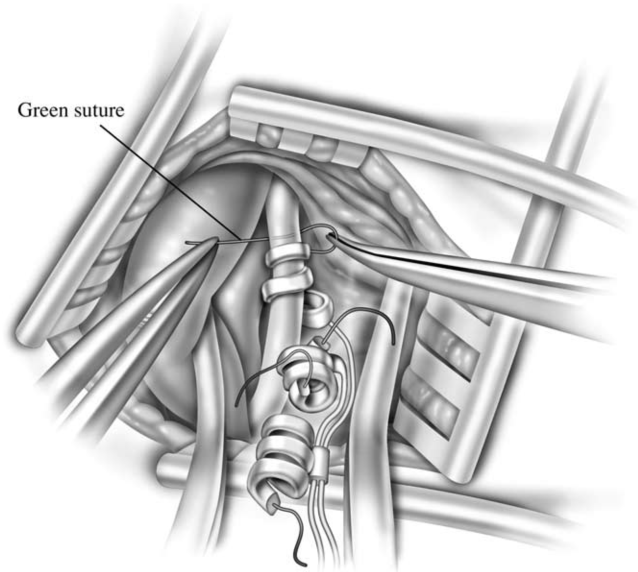

Figure 9 Figure demonstrates placement of the distal portion of the distal electrode (negative polarity, −) by controlling the passage of the embedded green suture under the vagus nerve from lateral to medial.

suture and pulling medially (perpendicularly relative to the nerve) while simultaneously grasping the most caudal green embedded suture of the same distal electrode (−) laterally. This maneuver allows the nerve to intercalate and rest within the middle portion of the spiral electrode (Figure 8).

The remaining distal end of the same electrode (−) is wrapped under from lateral to medial and then over and lateral to the vagus nerve by using the green embedded sutures (Figure 9). Similarly, the proximal end of the same electrode (−) is wrapped over and medial to the vagus nerve (Figure 10). Minimal retraction is used, and specifically no rubber vessel loops are used to pull the nerve out of the field; instead, the surgical dissection should provide adequate exposure to allow the surgeon to work in the natural plane of the vagus nerve.

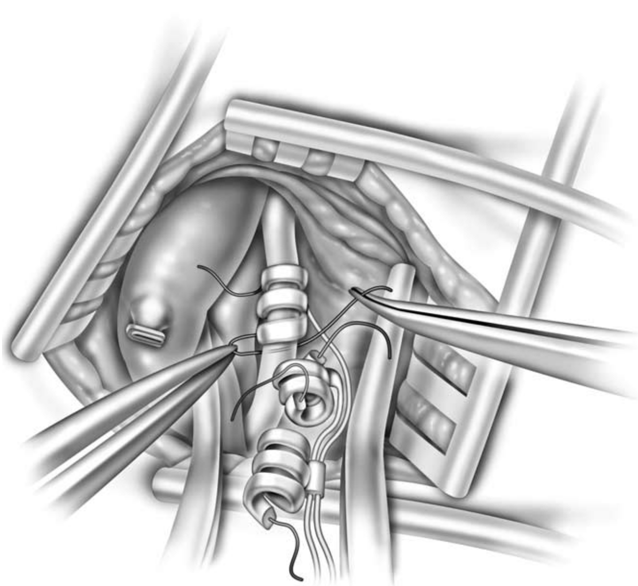

Figure 10 Placement of the proximal portion of the distal electrode (negative polarity, −) around the vagus nerve by controlling the passage of the embedded green suture over the nerve from lateral to medial and then under the nerve from medial to lateral.

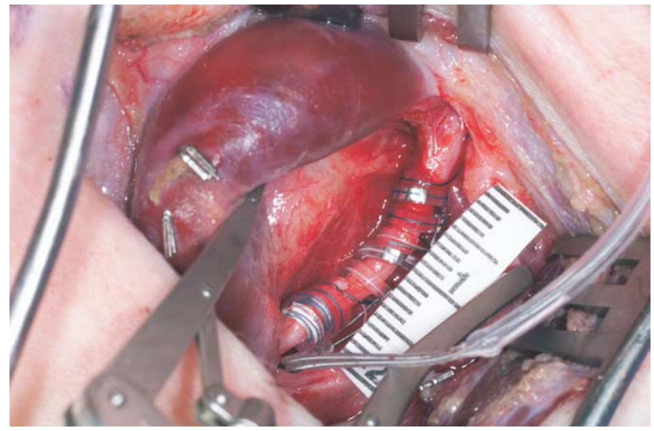

Figure 11 The helical electrodes in place on the vagus nerve without kinking or compression of the nerve (ruler shows millimeters). (Reprinted from Santos PM: Evaluation of laryngeal function after implantation of the vagus nerve stimulation device. Otolaryngol Head Neck Surg 129:269-273, 2003. © 2003 with permission from the American Academy of Otolaryngology—Head and Neck Surgery Foundation, Inc.)

The placement of the middle electrode (+) and the proximal inactive anchor tether is performed in an identical fashion. The two helical electrodes and the anchor tether should align with the nerve without kinking or compressing the nerve (Figure 11). The wound is copiously irrigated. A strain relief bend is created in the lead at least 1 cm below the anchor tether array to allow for the subsequent placement of a strain relief loop. The strain relief loop provides for adequate slack in the lead so that turning the head or muscular contractions does not pull the electrodes and vagus nerve. Tie downs are positioned and secured to fascia and not to muscle with 4-0 Vicryl sutures (Somerville, NJ) both deep and superficial to the sternocleidomastoid muscle (Figure 12). After the lead is placed, the alms retractor is removed. To avoid retracting the electrode array, the Weitlander retractors are carefully removed.

Operative procedure: Part 2. Placement of the VNS therapy pulse generator

The preaxillary incision is 5 cm in length. Sharp and blunt dissection is created over the pectoralis major muscle, thereby creating a pocket for the pulse generator. The surgical space should be slightly larger than the pulse generator. If the space created is too small, the lateral aspect of the generator will push against the incision site and cause potential breakdown of the layered wound closure. If the space is too large, a seroma may form and increase the chance of infection. The lead wire is passed subcutaneously from the cervical incision to the preaxillary incision with the assistance of the tunneler. The FDA-approved labeling describes tunneling with the blunt-tipped tunneler from the cervical to preaxillary incision. Specifically, the blunt-tipped tunneler is grasped with the right hand and carefully advanced in a subplatysmal plane and over the clavicle subcutaneously from the cervical incision to the preaxillary incision, with the left hand helping to direct the tip. Once the tunneler tip

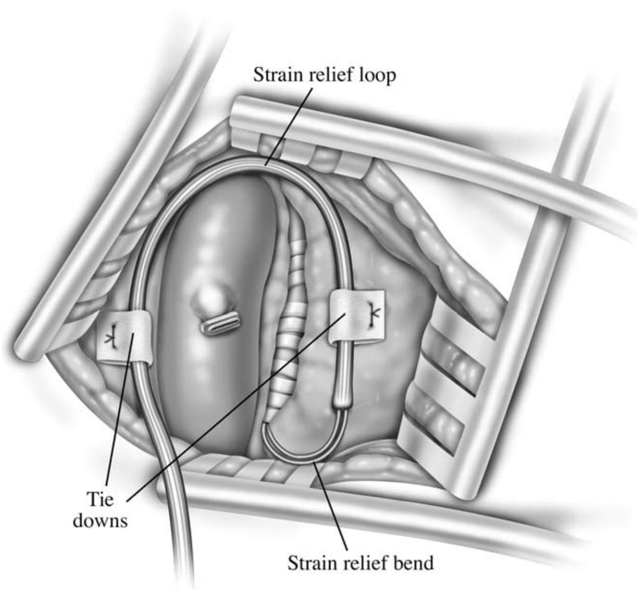

Figure 12 The strain-relief bend and loop with a "tie down" placement to prevent inadvertent pulling on the electrodes as the head turns, etc; note that the electrodes are covered with fascia to provide more protection.

is seen through the adventitia, a #15 blade can be used to incise the adventitia to allow passage of the tunneler. After delivery of the tip of the tunneler beyond the preaxillary incision (Figure 13), the tip is unscrewed and removed. The metal rod is removed from the plastic tubing, and the connector pin from the lead is passed into the end of the plastic tubing. The plastic tubing is pulled out of the preaxillary incision with the lead wire now in the preaxillary surgical space. Before placement of the connector pin into the pulse generator, the hex screwdriver is inserted into the setscrew and held perpendicular to the pulse generator, which allows back venting of the pressure resulting from connector pin lead insertion. The connector pin is placed deeply into the pulse generator and visualized through the clear casing to confirm complete insertion. The hex screwdriver is turned clockwise until it clicks, indicating a tight and adequate fit. A failed lead test during implantation is commonly caused by improper lead insertion.

The pulse generator and the attached lead are placed on the patient's chest and the first lead test is performed with the sterile sleeved programming wand. Testing should produce low impedance and thus confirm that the components are functioning properly through a closed circuit that includes the vagus nerve. After the lead test, the preaxillary space is copiously irrigated with saline, and the pulse generator is placed within the preaxillary space. A 4-0 Vicryl suture is passed through the plastic hole in the pulse generator and the pectoralis major muscle fascia. The surgical site should be irrigated copiously again with saline. A second lead test is performed with the pulse generator in the surgical site. Interrogation of the device should follow the lead test, and the device will be turned off.

Skin closure is performed by creating a watertight seal with a 4-0 Vicryl suture subcutaneously within the preaxillary space. Staples are applied to the preaxillary incision line. The cervical incision site is once again irrigated with saline and then closed in layers to avoid residual dead space and formation of seroma. The sternocleidomastoid muscle fascia is approximated to the strap muscle fascia with a 4-0 Vicryl suture. The platysma muscle is closed with a 4-0 Vicryl suture, thus creating a water-tight closure. The cervical skin is closed with a 5-0 nylon suture. Antibiotic ointment is applied to the two incision lines. No dressings or drains are used.

Discharge instructions

The patient is discharged to home and instructed to take oral antibiotics for 3 days, to use antibiotic ointment twice daily on the skin incisions for 2 days, and to avoid heavy lifting or being lifted under the axilla for 2 to 3 weeks. The patient is given the patient kit as supplied by the manufacturer and is trained by the neurologist in how to use the magnet. One week after the surgery, the patient is seen as an outpatient for wound check, suture and staple removal, and evaluation of voice and true vocal cord function. The patient should see the referring neurologist 2 weeks after the procedure for programming of the VNS therapy system.

Management of complications

Successful management of the complication of infection depends on early recognition and aggressive intervention. Early recognition depends on excellent patient education

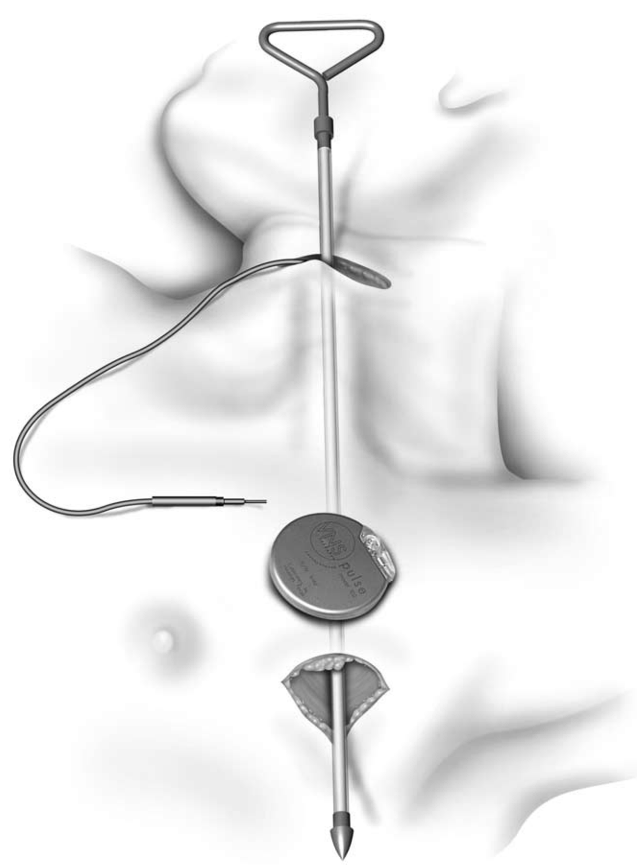

Figure 13 The tunneling tool in position to allow passage of the lead through the plastic sleeve subcutaneous tunnel for connection of the lead connector to the pulse generator.

and appropriate follow-up. Wound infections associated with implanted devices may occur within days or months of the procedure. Infections seem to occur more commonly with the axillary incision compared with the cervical incision.[21] Despite early intervention and aggressive intravenous antibiotics, many deep infections require removal of the device and replacement after the infection is controlled. A seroma without infection should be given time to resorb. However, if it does not resorb, it should be drained under sterile conditions.

The complication of left true vocal cord paresis/paralysis is best managed with recognition of this possible complication and careful monitoring of the laryngeal examination. This condition tends to improve over time. Early aspiration pneumonia, however, must be diligently treated and monitored.[2]

Conclusion

As the number of patients who require implantation of the VNS therapy device increases, the requirement for well-trained surgeons to perform the procedure will likewise increase. Electrode placement within the neck by otolaryngologists has been previously described within the head and neck literature as safe.[25] With appropriate training and planning on the part of the surgeon and the surgical team, implantation of the VNS therapy device is a relatively straightforward surgical procedure.

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